

Problem 10.2

If the constant force of interest is 6%, what is the corresponding annual effective rate of interest?

Solution

Let δ be the constant force of interest and let i be the annual effective rate of interest. The relationship between them is

$$\delta = \ln(1 + i).$$

Given

$$\delta = 0.06,$$

we have

$$0.06 = \ln(1 + i).$$

Exponentiating both sides,

$$e^{0.06} = 1 + i.$$

Thus,

$$i = e^{0.06} - 1.$$

Therefore,

$$i = 0.06183654.$$

So the corresponding annual effective rate of interest is

$$\boxed{6.183654\%}.$$